

The L^p norm and the σ hypothesis: what was wrong and what was done

Working note, Sigmoid_Universal_Approximation

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Abstract

This note began as an explanation of one extra hypothesis — σ assumed continuous in the L^p theorems, where Costarelli and Spigler assume only bounded sigmoidal. Investigating it turned up a deeper problem in the *definition* of the norm. Both are now fixed, in three stages: an integrability guard (§3), a move to the extended reals (§4), and the weakening of the hypothesis itself (§5). All four L^p theorems now assume only $\sigma \in \text{borel_measurable borel}$, and the Heaviside function witnesses that this is a real weakening rather than a restatement.

1 The original observation

Costarelli and Spigler state Theorem 3.2 as:

Let σ be a bounded sigmoidal function and let $f \in L^p[a, b]$, $1 \leq p < \infty$, be fixed. Then, for every $\varepsilon > 0$ there exists $N \in \mathbb{N}^+$ and a linear combination G_N of sigmoidal functions as in (2.1), such that $\|G_N - f\|_{L^p[a, b]} < \varepsilon$.

Our formalisation carried `assumes cont_sigmoidal`: "`continuous_on UNIV  $\sigma$` ", which the paper does not. The question was why.

2 Why: the junk-value collapse

The norm was originally defined as a bare Bochner integral, `Lp_norm p a b g = (∫ |g|^p)1/p`. Isabelle's `integralL` is a *total* function returning the junk value 0 on a non-integrable argument (`not_integrable_integral_eq`), and `powr 0` gives $0^{1/p} = 0$. So whenever $|g|^p$ failed to be integrable, `Lp_norm p a b g = 0 < ε` held for *every* $\varepsilon > 0$.

This is not hypothetical. "Bounded sigmoidal" constrains only the sup norm and the two limits, so with $V \subset (0, 1)$ a Vitali set,

$$\sigma(x) = 0 \ (x \leq 0), \quad \sigma(x) = [x \in V] \ (0 < x < 1), \quad \sigma(x) = 1 \ (x \geq 1)$$

is a bounded sigmoidal function in exactly the paper's sense and is not measurable.

Writing $\|G_N - f\|_{L^p}$ presupposes $G_N - f \in L^p$; on paper the notation carries that silently, in a total logic it must be discharged. The formalisation had discharged it with a hypothesis — but one stronger than necessary. The diagnosis was right; the remedy was the wrong shape.

3 Stage 1: the integrability guard

The defect is in the definition, so that is where it was fixed:

```
"Lp_norm p a b g =
(if integrable (lebesgue_on {a..b})) (λx. |g x| powr p)
then (∫x. |g x| powr p ∂(lebesgue_on {a..b})) powr (1 / p)
else undefined)"
```

and identically for `multivariate_Lp_norm`. These are internal real-valued working forms, not domain-enforcing norms: `undefined` is an ordinary unspecified real. An inequality involving it does not imply integrability. The public statements use `Lp_enorm`, which now returns infinity for a nonmeasurable powered absolute value. The verified lemma `Lp_enorm_finite_integrable` shows that a finite public error bound entails integrability. Uses of the internal real-valued forms retain explicit integrability premises.

It found real gaps. Theorems 3.2 and 5.4 each needed a genuinely new lemma — integrability of $|G_N - f|^p$ — that no proof had established, because the junk value had been absorbing it. Both follow from `Lp_Minkowski_integrable` on the splitting $G_N - f = (G_N - g) + (g - f)$ the estimates already use.

4 Stage 2: the extended reals

Guarding on integrability is sound but not honest: for measurable g with a divergent integral — $\int_0^1 1/x$ — the norm is not undefined, it is ∞ . So the norm the theorems are stated with is now `Lp_enorm`, `ennreal`-valued, guarded on *measurability*, generic over the measure. Its behaviour is proved: `Lp_enorm_eq_top` (divergent $\Rightarrow$ top), `Lp_enorm_eq_seminorm` (integrable $\Rightarrow$ agrees with the real-valued seminorm), and

$$\text{Lp_enorm_less_top_iff} : \quad \text{Lp_enorm } p \ M \ g < \text{top} \iff \text{integrable } M \ (\lambda x. |g \ x|^p),$$

the characterisation a real-valued norm cannot express.

The extended reals do not remove the guard. `nn_integral` is total, and on a non-measurable argument returns the *lower* integral, which can be finite and small. The guard survives but weakens from *integrable* to *measurable*. The two mechanisms compose: the extended reals handle divergence, the guard handles non-measurability.

Architecture. The four theorems are stated with `Lp_enorm`. Hölder and Minkowski stay in *real* on `Lp_seminorm`, where they are already proved and are used only *inside* proofs with integrability in hand; `Lp_enorm_eq_Lp_norm` and `Lp_enorm_eq_multivariate_Lp_norm` bridge the layers. Migrating `Lp_seminorm` too would have forced ~ 270 lines of Hölder and Minkowski to be reproved under truncated subtraction, for no gain in what the theorems say.

5 Stage 3: the hypothesis weakened

All four theorems now assume `meas_sigmoidal`: " $\sigma \in \text{borel_measurable borel}$ ". This suffices because $x \mapsto w(x-t)$ is continuous hence Borel, so σ composed with it is Borel; the network is a finite linear combination of such terms, hence Borel and bounded; and on a finite-measure set with $|f|^p$ integrable, $|G_N - f|^p$ is integrable.

The multivariate half was free. Six textual substitutions, built first try. `multivariate_network_measurab` had only ever used continuity in its first line, to derive measurability via `borel\_measurable\_continuous_onI`.

The one-dimensional half was not, and propagated further than expected. Dropping continuity of σ means the *network* is no longer continuous either, so two facts that had come free from “continuous on a compact interval” had to be rebuilt.

- Theorem 3.1’s proof is now measure-theoretic, mirroring Theorem 5.3: the network is Borel by a new reusable `sigma_shift_measurable`; integrability comes from `finite_measure.integrable_cons` against the uniform bound; the estimate from `integral_mono_AE`. The Henstock–Kurzweil bridge `Lp_norm_continuous_eq` is no longer on the path (it remains, as a fact about continuous integrands).
- **Theorem 3.2 was the unanticipated part.** Both its `G_meas` and its `Gg_int_ab` rested on continuity of the network, and integrability of $|G - g|^p$ cannot be recovered from an L^p bound — a small norm does not imply integrability. The fix was to strengthen Theorem 3.1 to *return* the integrability it already proves internally; Theorem 3.2 and the `ennreal` form of 3.1 both take it from there instead of re-deriving it.
- `borel_measurable_lebesgue_onI` was added to `Lp_Inequalities`: the general `euclidean_space` form of a helper that at the time existed only for `vec`, inside a mollifier theory that `Lp_Approximation` does not import. (That `vec`-only variant has since been deleted, and every call site now uses the general form.)

The witness. `Concrete_Sigmoidal_Examples.thy` now carries a third family beside the logistic and Gompertz sigmoids, both of which are smooth: `heaviside_is_bounded_function`, `heaviside_measurable`, `heaviside_not_continuous`, and `heaviside_Lp_approximation_theorem` — Theorem 3.1 instantiated at the Heaviside step function. That instance was *not provable* under the old hypothesis, which is what makes measurability a repair rather than an over-correction.

`sigmoidal_network_continuous` survives and still holds, but is no longer on the path to any theorem.

6 Practical notes worth keeping

- **$\top$ does not lex in this session.** `Lp_Inequalities` checked clean standalone under `-l HOL-Analysis` and then failed in the real build: the session’s parent is `Real_and_Complex_Analytic/Smooth` not plain `HOL-Analysis`. Use the constant name `top`. Corollary: *a standalone eval_at in a different logic is not evidence about this session.*
- **`ennreal` has no `powr` instance**, and no `ln` instance to build one on — hence `ennreal_powr`. Its minus is truncated ($\lambda a\ b. \max\ 0\ (a - b)$).
- **Conversions only go upward.** The bridges map `real` $\rightarrow$ `ennreal`, never back: `enn2real top = 0` is a `[simp]` rule, so converting down at the boundary would reintroduce the collapse being removed.
- **`\<^const>` in a header paragraph fails** when the paragraph precedes the definition it names. Three build cycles were lost to this; in a header, write plain `\<open>Name\<close>`.
- **Filter tactics diverged where an explicit ε - δ proof did not.** Proving `heaviside_not_continuous` via `filterlim_at_split` and `tendsto_cong` ran >20 min at ~ 5 GB, twice. Naming the δ explicitly and never constructing a limit terminates in seconds, and is half the length. Prefer the proof that gives a tactic no filter to search over.

- **A timed-out run reports zero errors.** `timeout` returning 124 means nothing was verified — a hang masks every proof after it. Always check the exit code, not just the error count. Also, `timeout` signals the `isabelle` wrapper and not the `poly` child, which survives and spins at 100% CPU; kill it before starting anything else.